

Examining the Institutions in Enabling the Effective Management of Biosphere Reserves: A Case Study of the Biosphere Reserve of the Apayao (BRyA)

Selected Excerpts from Undergraduate Thesis

Biodiversity Governance and Institutional Analysis

This document contains selected, anonymized excerpts from the author's undergraduate thesis. All sensitive information, personal identifiers, and confidential data have been removed.

Readers may wish to focus on the Results and Discussion section (pp. 16–25), which presents the study's core findings on multi-level biodiversity governance.

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Research Context and Governance Framework (Excerpts from Chapter 1)

Introduction to Biosphere Reserves

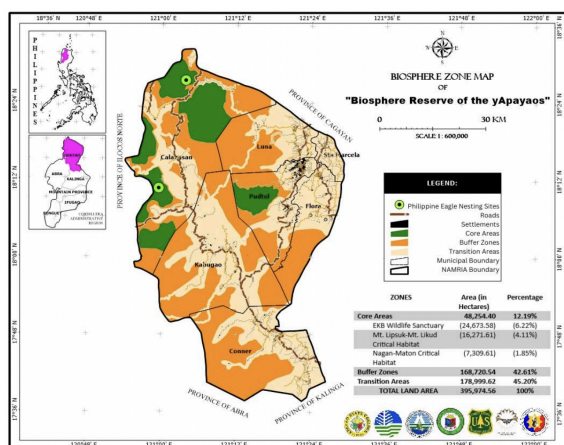
The United Nations Educational, Scientific and Cultural Organization (UNESCO) Biosphere Reserve (BR) designation, the lesser-known counterpart of the World Heritage Site designation, can be examined as a way to answer this question of reconciliation between progress and environmental protection. Defined as “areas of terrestrial and coastal/marine ecosystems, or a combination thereof” recognized under the UNESCO's Programme on Man and the Biosphere (MAB), BRs integrate learnings from natural science with the social sciences to nurture sustainable, culturally appropriate, and environmentally friendly development (UNESCO, n.d.; UNESCO, 2013).

BRs in the Philippine Context

There are currently four biosphere reserves in the Philippines; namely, the Albay Biosphere Reserve, the Palawan Biosphere Reserve, the Puerto Galera Biosphere Reserve, and the Biosphere Reserve of the yApayaos (BRyA). The province of Apayao, which was only designated as a biosphere reserve last July 5, 2024, is a 3rd income class province in the Cordillera Administrative Region (CAR) with seven (7) municipalities and a total population of 124,366 as of the 2020 census (Philippine Statistics Authority, n.d.).

Figure 4

Zonation Map of the Biosphere Reserve of the yApayaos



The map above shows the zonation map included in the nomination dossier of the BRyA. Core areas here are colored solid green, buffer zones are colored solid orange, and transition areas are colored solid light orange. Settlements are colored solid black while boundaries and roads are symbolized by black lines of varying widths. It is worth noting that most of the core areas are situated in Calanasan, Apayao, with the remaining in Pudtol. In addition, it can be observed that multiple small-scale core areas are separate from one another. This type of biosphere reserve zonation is referred to by UNESCO (2013) as a cluster biosphere reserve zonation system.

RESEARCH QUESTION

Given the context of UNESCO Biosphere Reserves and the province of Apayao, the research question of this study is as follows: “What are the institutions and processes of the Biosphere Reserve of the yApayaos (BRyA) in the province of Apayao and how do they enable its effective management?” This question has the primary objective of investigating how environmental, political, economic, cultural, social, or educational institutions and processes related to the BRyA are structured and implemented, in addition to examining how the province’s context might have influenced the way they were developed. The question also examines how these institutions can contribute to the sustainable and effective management of the biosphere reserve in the long term. According to Russell and Harshbarger (2003), “Institutions span economic, social, religious, and political life and provide the framework within which people act and make choices.” This includes laws, policies, management structures, cultural traditions and practices, and other social norms related to the BR and the overall management of biodiversity and natural resources within the province.

Research Objectives

In addition to the primary goals of description and evaluation, the research question also has the following objectives:

- 1) To examine whether or not the designation processes and goals appropriately fit the socio-ecological context of the communities in the province,

- 2) To recommend changes or transformations needed to ensure that the BR management can accrue positive outcomes for the biosphere reserve, and
- 3) To identify the local and cultural mechanisms that can contribute to the effective management of the BR.

SCOPE AND LIMITATIONS

Following the research question, the scope of this research study was only limited to the province of Apayao and the local and national political, economic, cultural, social, educational, and environmental institutions and policies it is situated in specifically related to the management of the BRyA. As such, managing authorities who were interviewed were only limited to the offices or organizations stated in the Biosphere Management Council of Apayao (BMCA). It also only examined the indigenous knowledge and practices related to environmental management specific to the indigenous cultural communities (ICCs) within the province of Apayao strictly through a secondary review of literature. While the research study focused on the UNESCO Biosphere Reserve as a site designation, it also examined other protected area designations under the National Integrated Protected Areas System (NIPAS) Act, as this zonation system requires the inclusion of legally protected areas as core zones.

EMPIRICAL AND THEORETICAL SIGNIFICANCE

The research question posed above addresses an empirical gap in the body of research regarding conservation management and practices, specifically in and unique to the province of Apayao. For instance, there exists a postgraduate study by Cayaba (2018) regarding the *lapat* system or the indigenous natural resource management practice within the province. However, much more research needs to be conducted to achieve a holistic understanding of how environmental management is done and received by local communities within Apayao. Aside from adding to a very limited pool of knowledge in Philippine research, this study also opens opportunities to examine conservation mechanisms in other indigenous communities within the Cordillera Administrative Region. While there have been studies on BR management, including community participation and

even the creation of biocultural community protocols to aid in the nomination process, there is still a need for more research on the experiences of stakeholders during the early-stage designation of BRs (Agama et al., 2011; Jaafar et al. 2023). Examining the role of indigenous knowledge and practices can also reveal significant insights about how indigenous communities in the country can contribute to environmental management and protection. In addition, this study focuses not on the outcomes of effective BR management but on the institutions, practices, and other factors that can enable it, which Plummer et al. (2017) affirm by stating that “incorporating such variables into future research will enhance the ability to connect evidence from decisions through to impact.” In this way, the study also offers practical and real-life evaluations and examples of institutional mechanisms that may be replicated in other protected areas, conservation sites, or biosphere reserves within the Philippines.

Review of Related Literature (Excerpts from Chapter 2)

POLITICAL INSTITUTIONS IN BR MANAGEMENT

Significance of Governance Structures in BR Management

Examining the political institutions that enable BR management also means examining the governance structures and principles on which different management approaches are founded. At the core of “enabling” political institutions in BR management is good governance that is appropriate and contextualized to the site’s socioeconomic and geographical location. The Statutory Framework of the World Network of Biosphere Reserves (1996) requires each BR to have a designated authority or mechanism to implement a management policy or plan of the site. Governance structures of BRs determine the extent of participation of stakeholders in decision-making, inform the development approaches that are employed to meet the unique goals of the BR, and ultimately set the foundations for the preparation, implementation, and monitoring of the site, including the political buy-in needed to support it (Schliep & Stoll-Kleemann, 2010; UNESCO, 2022; Klaver et al. 2024).

Heterogeneous vs. Multi-Stakeholder Management Approaches

The Management Manual for UNESCO Biosphere Reserves in Africa, edited by the German Commission for UNESCO (2015), categorizes the “authority model” and the “NGO model” as the two most common or ideal governance models for BRs in Africa. The authority model, which can also be seen to be discussed by other literature as “government management” or “heterogeneous management,” describes a top-down model where management is centralized and dependent on the national or provincial government (Schliep & Stoll-Kleemann, 2010; Amer et al., 2015; Tarasco, 2020). BRs under this model are also described as placing too much priority on *in situ* conservation in core areas, neglecting sustainable development in buffer and transition zones, and oftentimes failing to strictly enforce laws. The NGO Model, in comparison, is described as employing “multi-stakeholder” approaches as an independent authority that still observes vertical power-sharing but at the

same time highlights horizontal collaboration between a diverse stakeholder group (Amer et al., 2015; Hedden-Dunkhorst & Schmitt, 2020; UNESCO, 2022).

While both models necessitate multi-stakeholder collaboration and participation of “a suitable range of *inter alia* public authorities” to some extent as required by the Statutory Framework, BRs that are not dependent on local authority and instead rely on a governing body composed of different institutions are seen to have better development in all zones and overall better coordination, especially concerning conflicting interests, which is a common challenge across many BRs around the globe (Batisse, 1982; UNESCO, 1996; Amer et al. 2015; Taggart-Hodge & Schoon, 2016; Tran et al. 2019; UNESCO, 2022). BRs whose management authorities highlight and prioritize multi-stakeholder and participatory approaches are also regarded within the literature as better suited to BR management especially because of the paradigm shift from strict biodiversity conservation to overall sustainable development and capacity-building, which is reflected in the goals of the Man and the Biosphere Program (Ferreira et al. 2020; UNESCO 2022).

Challenges of “Fragmented” and Multi-level Management

It has been established within the literature that the management structures that prioritize participatory decision-making and multi-stakeholder collaboration are seen to be the more “enabling” tools for effective BR management compared to BRs that have centralized authority. However, there are many challenges in effectively implementing and organizing “fragmented” or multi-level management. Among these is the tendency for multi-stakeholder management to complicate “the already complex layering of management authority, laws, and zones” and the reconciliation of “local and global objectives of conservation” (Berkes, 2007; Dressler & McDermott, 2010; Schliep & Stoll-Kleeman, 2010). Shifting to a sustainable development paradigm from a strict *in situ* conservation approach also implies an added difficulty in managing trade-offs between rural development and conservation, especially for larger sites that are being handled by multiple organizations and institutions, which can compromise ecological integrity and foster overexploitation of resources (Batisse, 1982; Ma et al. 2009; Sinamai, 2014). In addition, literature also suggests that the impacts of participatory mechanisms in BR management should not be viewed through a normative lens; for example, communities in BRs in Russia, Indonesia, and

Bangladesh experience some degree of capacity building and multi-stakeholder collaboration but see low levels of community willingness to participate due to other factors such as livelihood disruptions or leadership conflict (Jaafar et al. 2023). In addition, Batisse (1997) notes that the nature of conservation efforts and where they are focused “depends on what objective is being pursued”, which can vary from site to site depending on the specific goals of management bodies. These challenges and nuances in examining BR management authorities are important to consider, as what is considered “enabling” varies across different geographic and socioeconomic contexts.

The National Integrated and Protected Areas System (NIPAS) Act of 1992

In the Philippines, multi-stakeholder approaches to biodiversity conservation can be observed through the National Integrated and Protected Areas System (NIPAS) Act of 1992, a law that establishes a framework for decentralized, community-based reserve management within the country (Republic Act No. 7586). This legislation standardizes collaboration across conservation experts, scientists, citizens, indigenous communities, and local government units while situating local volunteer groups as “vital links” in the protection efforts of legally protected areas (Senga, 2001). In this way, the governance structure and management model that this act provides can be likened to the “NGO model” of BR management, wherein great importance is placed on multi-stakeholder collaboration. However, it differs from other bottom-up managed BRs in that it also requires the participation of the Department of Environment and Natural Resources (DENR) as a monitoring and evaluation body for all legally protected areas within the country.

By leveraging on the value that indigenous communities place on their indigenous traditions, indigenous communities gain a sense of autonomy and ownership of conservation, increasing self-determination and empowering them to protect their daily activities and forms of livelihood, which are prone to be deemed illegal by the state (Perez, 2018). Areas not under the NIPAS Act may make it more difficult to generate political buy-in and political pressure to designate new sites as BRs, especially as support for the concept relies on “popular political benefit” (Brown, 2002; Coetzer et al., 2013). In addition, the BRyA is unique in that no other BR within the literature employs the use of the *lapat system* of the Isnag community as a tool to increase compliance with biodiversity conservation and

protection mechanisms. This has many cultural implications for the BRyA, which will be discussed in the latter part of this review.

ECONOMIC INSTITUTIONS IN BR MANAGEMENT

Economic Challenges of BRs

It has been established that sustainable development initiatives within BRs, such as eco-cultural activities, can enable effective BR management in many ways. However, due to the complex multi-stakeholder approach of the BR as a concept, many economic trade-offs and challenges need to be considered. Firstly, the economic development of the buffer and transition zones within the BR implemented without an “ecosystem-based approach” may lead to over-exploitation and the loss of ecological integrity (Batisse 1982; Ma et al. 2009). Secondly, a key economic issue within BRs is the issue of resource allocation and access across stakeholders. Spatial and resource restrictions due to the zonation system can negatively impact the livelihoods of local communities, which then raises the issue of just compensation, displacement, and a decrease in household income (Coetzer et al. 2013; Ferreira et al. 2020; Jaafar et al. 2023). This issue of managing costs is also an issue among management authorities, especially in larger BRs. For example, an article about the BRs in Japan by Tanaka & Wakamatsu (2018) observes that “multi-municipal cooperation produces huge transaction costs.”

CULTURAL INSTITUTIONS IN BR MANAGEMENT

Local and Traditional Knowledge in BR Management

It is essential to discuss the cultural institutions that enable the effective management of BRs, as many sites encompass local communities with unique cultures and traditions. A concept integral to understanding the role of local communities and indigenous peoples in BRs is “co-management”, which is an approach for protected area management that is characterized by “shared responsibilities and joint decision-making of key stakeholders” (Berkes, 2008; Schultz et al., 2010; Major et al., 2018). This concept used in environmental management is described to be adaptive and evolving through a process called “social learning,” which allows it to become a dynamic process informed by iteration

and collaboration (Berkes, 2008). Understanding co-management means identifying local and indigenous communities as “co-managers” who are integral to the management process and whose involvement is actively sought (Major et al., 2018). This also means recognizing the value of their local knowledge and traditions in biodiversity and ecosystem conservation and integrating it into management approaches (Schultz et al., 2010). In the BRyA, the *lapat* system ordinance leverages the cultural value of the indigenous tradition to increase compliance and participation in conservation and protection laws. This way, Isnag communities become co-managers by lending their knowledge to unify and involve communities in biodiversity conservation.

Challenges to Integrating Indigenous Knowledge in BR Management

The challenges involved in integrating indigenous knowledge and BR management are intricately tied to the political and economic arrangements of the site. Local communities need the resources to “facilitate indigenous leadership to shape external mechanisms for protected area establishment and development” (Tran et al. 2019). In addition, co-management is “weak” in reducing poverty and marginalization within local communities as decentralization often reinforces dominant and elite power structures within the site (Berkes, 2008; Dressler and McDermott, 2010). There is also an added difficulty of strengthening the legitimacy of co-management, especially in communities that are diverse and therefore have many conflicts of interest that need to be settled and reconciled. There is also a need to establish systems and processes that determine the best types of approaches to protected area conservation and management across different types of cultures and traditions, especially for communities that do not necessarily appear to “be living in harmony with nature” and therefore can be dismissed as inauthentic (Perez, 2018). As previously mentioned, biodiversity conservation and management should not be founded on normative narratives and principles, especially for indigenous communities whose values and beliefs are incredibly nuanced and require the proper education and awareness to understand.

SOCIAL AND EDUCATIONAL INSTITUTIONS IN BR MANAGEMENT

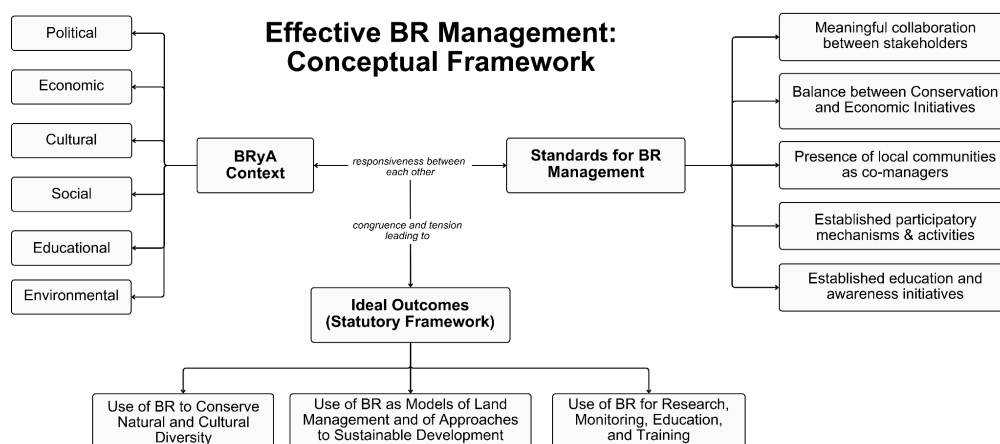
Importance of Participatory Approaches in BR Management

As the BR necessitates horizontal collaboration between different levels of organization within the site, participatory approaches become essential to bridging and reconciling the gaps between stakeholders. There is a consensus within the literature that participatory approaches and processes are essential to effective BR management, which is also reflected in the goals of the Man and the Biosphere (MAB) program and differentiates the BR concept from other protected area designations (Coetzer et al. 2014; Bouamrane et al., 2016; Ferreira et al. 2020; Leguia-Cruz, 2024). However, there needs to be a close examination of how specific communities participate and engage in activities related to the BR depending on their contexts. Understanding participation in the context of BRs necessitates determining the extent to which citizens value the site. The abstract values that communities place on protected areas have a direct correlation with the extent to which locals are willing to be involved in and participate in initiatives related to BR management.

CONCEPTUAL FRAMEWORK

Figure 5

Conceptual Framework for Effective Biosphere Reserve Management



The conceptual framework above, grounded in the findings of the review of related literature, details the relationships between three main concepts: the context of the BRyA, standards for BR management as seen in literature, and ideal outcomes of the BR as a

designation on the site as stated in the Statutory Framework. This conceptual framework is meant to guide the implementation of the methodology, which is discussed in the succeeding chapter. The elements under the “context of the BRyA” follow the structure of the review of related literature, in which the political, economic, cultural, social, educational, and environmental institutions of the BR site are examined.

Research Design and Methodology (Excerpts from Chapter 3)

RESEARCH DESIGN

As reflected in the previously discussed review of literature, while there can be objective or normative “standards” for what is considered effective BR management, the types of institutions that enable it are dependent on the specific context and background of the site. The question that this study then seeks to answer is how the institutions specific to the province of Apayao can enable the effective management of the BRyA. This particular research study employed an “exploratory” type of case study where the object being studied has no clear, single set of outcomes (Baxter & Jack, 2008).

DATA GATHERING METHODS

For this research study, the research methodology was grounded in Rapid Rural Appraisal (RRA) techniques. According to Russell & Harshbarger (2003), rapid rural appraisals that employ purposive sampling are used to produce rich information about “populations within a conservation area that encompasses several ethnic groups and resource management systems.” Following this, the primary data-gathering method used for the study was semi-structured key informant interviews (KIIs).

Direct observation, transect walks around government-proclaimed *lapat* areas in Pudtol, and auto-photography were also employed to triangulate the findings from the key informant interviews and focus group discussions. Furthermore, findings were constantly reviewed alongside the nomination dossier, relevant UNESCO documents, and the management plan to ground the findings in the conceptual framework and context of the biosphere reserve.

RESEARCH INSTRUMENTATION

As previously mentioned, the research study employed semi-structured interviews as the primary data-gathering method, which was guided by a “checklist” or a set of guide questions that were used but were not strictly adhered to. The conceptual categories under the social-ecological systems framework by Ferreira et al. (2018) were used to guide the creation of these guide questions (See Appendix A). The conceptual framework developed in

the review of related literature was also used to categorize and develop these guide questions. While the interview questions were created in English, I shifted to Ilokano depending on the preferences of the interviewees. These questions focus on gathering context and knowledge from the interviewees on the processes, challenges, and mechanisms related to the province's biosphere reserve management plans.

DATA ANALYSIS METHOD

After conducting the interviews, the data was encoded and transcribed in preparation for analysis. As the primary research design for this study was a case study design, the data analysis method used was thematic analysis of interview results. After conducting the interviews, I identified specific themes observed within the data following the indicators stated in the conceptual framework previously established in the review of the literature. If any gathered data did not align with the conceptual framework, the analysis then shifted to an inductive approach, where themes were taken *in vivo* from the data. Thematic analysis aided in revealing insights about how the knowledge of the different stakeholder groups aligns and harmonizes with each other. As previously mentioned, research volunteers were identified to aid in mitigating any biases in gathering and analyzing data. However, intercoder reliability assessments were no longer conducted as research volunteers were not able to attend all of the interviews. This is a significant limitation identified by the study that I recognize may have affected the interview process.

Results and Discussion (Excerpts from Chapter 4)

**A majority of the quotes have been redacted for privacy*

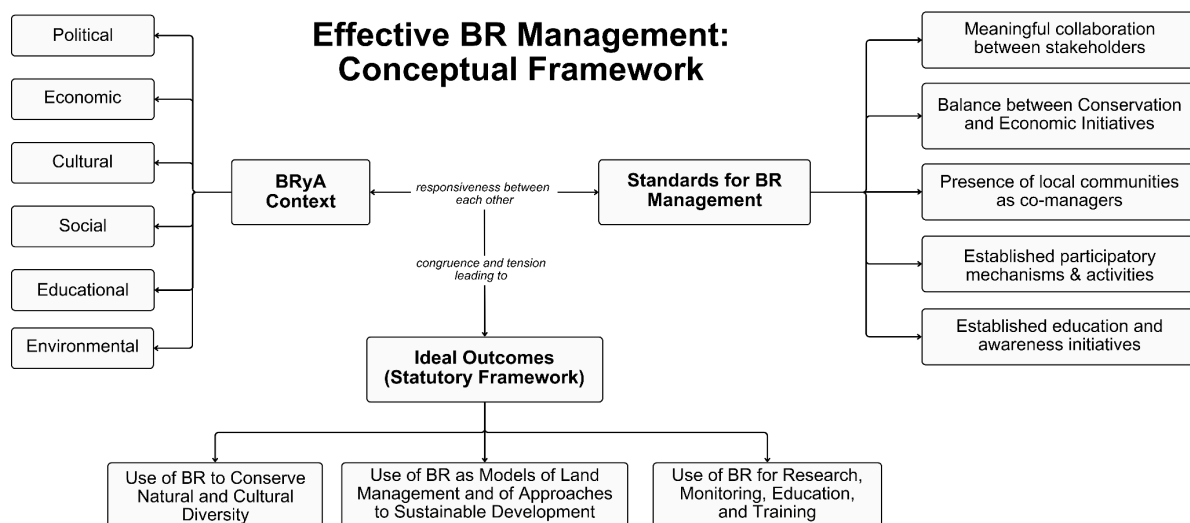
INTRODUCTION

Data Analysis Methodology

The analysis of data for this study is anchored on the research question “What are the institutions and processes of the Biosphere Reserve of the yApayaos (BRyA) in the province of Apayao and how do they enable its effective management?” As the question has the primary objective of investigating how environmental, political, economic, cultural, social, or educational institutions and processes related to the BRyA are structured and implemented, the findings of the study are categorized according to the aforementioned dimensions. However, as the coding process was inductive, codes may belong to more than one dimension or category. The creation of themes from the code groups was also informed by the study’s conceptual framework seen in the figure below.

Figure 8

Conceptual Framework for Effective Biosphere Reserve Management



This chapter is also divided into two categories, namely, results and discussion for findings from secondary literature and results and discussion for findings from qualitative interviews.

The coding process for the interview data, categorized by dimension according to the research question, generated a total of twelve themes as shown in the codebook tabulated below:

ENVIRONMENTAL THEMES

Integrated and Diverse Range of Ecosystems

The first, and possibly most important criterion for a biosphere reserve area to be qualified is that “it should encompass a mosaic of ecological systems representative of major biogeographic regions, including a gradation of human interventions” (UNESCO, 1996). A salient theme that emerged within the interview data is the general recognition and acknowledgement among the government officials of the high ecological value of the province of Apayao, from its large forest cover and watershed areas to its diverse range of flora and fauna.

The inscription of the province as a biosphere reserve was also largely attributed by the interviewees to the presence of an active nesting site of the *Pithecophaga jefferyi* or Philippine Eagle. For the province, specifically, the Eagle acts as a unifying symbol for the province’s cultural and ecological identity. The presence of the endangered species is regarded as indicative of the pristine conditions of the province’s forests:

“We were qualified because of the Philippine Eagle - these critical habitats are where the wildlife live, they are what support symbiotic relationships, and without them, our forests would degrade.”

The endangered species, which is quoted by one participant as an “outstanding universal value” of the province, was also the primary reason that the Philippine Eagle Foundation, one of the key stakeholders in the planning and nomination process of the BRyA, established a partnership with the provincial government in 2014. Since then, the foundation has aided in garnering support and increasing exposure of the conservation efforts of the province to the general public.

Figure 13

Philippine Eagle Nariha Kabugao in flight after release on April 12, 2024 (Erwin Mascarinas)



The presence of the Philippine Eagle also has significant political implications. The provincial government has worked closely with the Philippine Eagle Foundation (PEF) to nominate the province as a biosphere reserve since their involvement in the early 2010s. PEF has a critical role in the management and monitoring of the species and its nesting sites within the province. This means that power and influence regarding decisions for the management schemes of forests become centralized to the provincial government rather than to national line agencies or indigenous protected area management bodies. These are implications that will be discussed further in the chapter.

Complex Range of Resource Access Mechanisms

How the province mobilizes its environment and natural resources is also complex and diverse, in that there is a wide variety of environmental protection and conservation mechanisms utilized by the province. They can be categorized into either (1) protection mechanisms as managed and established by the provincial or municipal offices of the Department of Environment and Natural Resources (DENR), or (2) indigenous protection systems as established by local IP leadership or by the National Commission for Indigenous Peoples (NCIP). The institutionalization of the indigenous mourning practice *lapat*, which will also be discussed in the later sections of this chapter, is unique in that it falls in between, or rather overlaps, with both categories. The same can be said for the institutionalization of the *bantay gubat* or the green guard, which is a position offered to local communities by MLGUs to monitor activities within local forests. While there are a vast number of protection mechanisms as stated in the BRyA's management plan, a significant number of interviewees

specifically highlighted the establishment of critical habitats within the province, which served as the basis for the initial nomination of the province to be designated as a biosphere reserve.

POLITICAL THEMES

Concentration of Power and Interest within Provincial Government

The implications discussed in the previous theme concerning the consequences of diverging from the priorities of the provincial government can already be seen in the preference of a majority of officials in provincial-led management of protected areas (such as conservation via *lapat* or mechanisms under the IPRA Law) rather than DENR-led management (such as mechanisms under the PAMB). Most of the sentiment towards DENR-led management schemes and enforcement in general is that of either (1) weak enforcement or (2) a lack of political continuity among its members, especially since appointed national line agency employees have limited tenure.

Furthermore, Botha (2023) observes that there is a “link between high place attachment and the desire to protect the environment or natural resources.” While this observation was made about citizens and their beliefs in environmental protection, this can also be said for the officials of the provincial government. As previously mentioned, many of these officials are from rural indigenous communities themselves, most of which relied on the environment for livelihood and survival.

This perceived value of belongingness and ownership of the environment as something to be passed down from generation to generation of yApayaos can be observed to contribute to the perception of DENR-led management as unfit or lacking for the context of the province. To the interviewees, to be from the province of Apayao is equivalent to having a deeper sense of value for the land.

While literature on the standards for effective management of biosphere reserves notes the significance of locally-led management schemes in contextualization, Schliep & Stoll-Kleeman (2010) also note that BRs must receive “active support from the superordinate governance levels.” The preference toward strictly provincial-led management may affect how the management council establishes networks and linkages across varying levels and

forms of organization in the future, many of which will be essential for the sustainability of the biosphere reserve, especially in terms of resource allocation.

Necessity to Improve Intragovernmental Coordination

For the case of the province of Apayao, a homogenized political prioritization across the provincial government does not necessarily translate to better intragovernmental coordination. The current organizational structure of the Biosphere Management Council of Apayao and the Technical Working Group decentralizes core functions of the BR (conservation, development, and logistic support) across all line agencies and local government entities; at the same time, there are plans in place to create a separate BR management arm under the Provincial Government Office - Environment and Natural Resources Department (PGO-ENRD). While the approach is congruent with biosphere reserve standards in that the management is multi-stakeholder in nature, there needs to be better synergies and delineations of tasks among all stakeholders.

In addition, the functions and mandates of the DENR within the province are decentralized across multiple levels of government, from the provincial level to the municipal level. This means that there are multiple ENR officers (PENRO, CENRO Calanasan, CENRO Conner, MENRO) across different levels with different jurisdictions (provincial, upper Apayao, lower Apayao, and municipalities, respectively) who are in charge of implementing and monitoring DENR activities. This level of fragmentation can create difficulties in terms of coordination among stakeholders.

This issue is further compounded by the complex land titling schemes within the province, many of which are not strictly enforced enough, such that land owners are still able to negotiate terms on land use. This issue of land use can also be attributed to a fragmented management regime where government functions overlap. Land titling and land management within the province can be facilitated by multiple national line agencies, such as the DENR (for titles such as alienable and disposable land) and the NCIP (for the awarding of certificates of ancestral domain titles and claims). Contestation regarding the most appropriate land use and titling for certain areas can lead to claims over resources, especially considering the pre-existing land disputes and claims across all seven municipalities on the local government level. Dressler & McDermott (2010) observe the

same challenge in Palawan, noting that decentralization of environmental management in the province “further complicated the already complex layering of management authority, laws, and zones.”

ECONOMIC THEMES

High Tensions between Sustainable Development and Ecological Conservation

Literature shows that one of the main challenges in BR management is reconciling and balancing sustainable development goals with environmental protection and conservation initiatives. The same challenge can be observed in an emerging issue within the province that was discovered among the interview data, which is the increasing difficulty of municipalities to control solid waste management behaviors among communities, given the rising population of tourists. This is important to discuss because the biosphere reserve designation greatly increases social exposure to neighboring provinces, which can lead to an increase in visitors. The sheer number of tourists then makes it more difficult for municipalities to control and manage the amount of waste generated.

Key to the biosphere reserve designation is a “view to improving human livelihoods” (UNESCO, n.d.). Concerning losses of livelihood due to restrictions in spatial use and resource access, a significant portion of the interviewees noted the agricultural practice of *kaingin* or slash and burn agriculture as one of the most significant challenges facing the management bodies in the future. Interview data shows that there is an acknowledgement among all of the stakeholders of the management body that enforcing a strict restriction on *kaingin* on local communities is not possible because of the severe loss of livelihood that such a restriction can cause. There is a certain recognition that there are no viable alternative sources of livelihood, especially for communities in forested areas:

“Problema mi tatta ket kasano i-iwas dagitoy kaingin, diay uma. Haan mi nga mapaserdang ejay, isu nga adda etuy kaingin ordinance mi. Tapno lang atleast kuma ket ma-regulate, haan dan nga aguma ken dagijay closed forest. Kasi source of income da dagijay eh, source of food and livelihood.”

[Our problem today is how we’re going to avoid these kaingin, these uma [slash and burn]. We can’t stop them, which is why we have the ordinance. So at least we can

regulate it, so that they don't [practice] slash and burn in the closed forest[s]. Because that is a source of income, source of food and livelihood.]

This agricultural practice is particularly common in upland forest communities because the percentage of alienable and disposable land allowed for resource and livelihood use is only ten percent. The agricultural practice also has deep roots within the indigenous communities within the province because of its origins in indigenous culture. Although the practice is common among all municipalities, *kaingin* is only regulated in the municipality of Calanasan, wherein farmers are allowed to continue the practice provided that they do not expand to other sites. Although an inventory of existing *kaingin* sites currently exists within the municipality, many farmers are hesitant to register with the government due to the fear of being restricted or arrested:

"Diay Calanasan, the most asked question idiy ket kaingin. [...] Diay question da talaga about kaingin ket apay awal. Hesitant da nga talaga nga maiparit ti kaingin. Diay kunkunami nga ma-inventory ti kaingin, medyo mabuteng da nga ibaga ta, "baka maibalud kami," kuna da."

[In Calanasan, the most asked question is about kaingin. Their question about kaingin really is about why it's not allowed. They are hesitant because kaingin will be restricted. When we tell them that kaingin [sites] will be inventoried, they are scared to tell us where because, "we might be put in jail," they say.]

CULTURAL THEMES

Rich and Diverse Cultural History

What is arguably the most important cultural institution of the province of Apayao related to the management of the biosphere reserve is the cultural practice of *lapat*. The practice has great historical significance and is sacred to indigenous communities in the province, particularly that of the Isnag. Traditionally, the practice is used to venerate the dead, with the family of the recently bereaved restricting all forms of entry and use of a portion of land, river, tree, or other natural resources. This practice is especially significant to the discussion of effective management of the BRyA because it is perceived by many of the interviewees as central to the designation of the province as a BR because of its uniqueness in the local context. Similar to how the management bodies leveraged the presence of the Philippine Eagle nesting sites for the nomination, the cultural practice was also highlighted

to make the province distinct from existing protected areas and biosphere reserves within the country. In the context of the effective management of the BRyA, the integration of the rich and diverse cultural context of the province into the management regime reflects how well it is contextualized to the needs and backgrounds of local communities.

Andaya et. al. (2008) posit that “forest resources rest on a dynamic interplay of culture, power, history, and human agency.” As the management regime of the province is heavily influenced by its cultural context, the way the management interprets shared meaning in forest protection and conservation becomes very significant. This is important, especially because there is an existing perceived difficulty of “uniting” cultural identities in the province. While the majority of the population is of indigenous representation, the nuances of IP culture and IP practices can widely differ from municipality to municipality. Cultural contexts can be very different within Isnag communities themselves, with delineations being made between upstream and downstream communities, for example. There is a large range and spectrum of cultural contexts, and thus a difficulty with finding a “unified” identity that can be used as “branding” for the BRyA.

Significant Integration of Cultural Mechanisms in Governance

It is important to mention that the aforementioned cultural practice of *lapat* was integrated into environmental governance in the municipality of Calanasan via Municipal Ordinance No. 12-2011, series of 2011. This had significant implications for the meanings and values behind the tradition. The formalization of the tradition into law was observed to be met with a degree of social hesitance and opposition by local communities due to the change in the use and goals of the practice from that of grieving and mourning bereaved family members to environmental conservation and protection. It can then be said that institutionalizing the tradition facilitated a shift in the meaning and value of the sacred practice. Several participants echo the apprehension of members of the community:

“Dayta lapat system nga practice ti Isnag, dati dati ti original na dayta ket it's a way of mourning lang when somebody dies in the family. Dagituy nagka-una, haan da ngamin pinanpanunut nga on the other hand, adda gayam pagsiyaatan na dagidiay.”

[This lapat system which is a practice of the Isnag, before, originally, it's a way of mourning for when somebody dies in the family. But these [elders], they are unable to think that on the other hand, there are other benefits to those practices.]

A common sentiment that the interviewees had regarding the problematization of this formalization was that the change in meaning was necessary in lobbying for the policy as a contextualized and grassroots mechanism for environmental conservation. Straying from the original intentions and uses of the cultural practice was then seen by many of the government officials to be a necessary trade-off to maximize its indirect ecological benefits.

This perspective from the government officials has significant implications for the way cultural discourse in the province is shaped. While there is a recognition and acknowledgement of the sacredness of the practice in its traditional form, the benefits of institutionalizing the practice are regarded as outweighing any negative impacts the change in meaning may have on communities. Apart from its potential as a tool for ecosystem regeneration, institutionalizing the practice was also leveraged as a way to prevent future conflicts on land ownership and land claiming, which was referenced by participants as common challenges with the use of the traditional practice. There is a need to examine whether social opposition to the ordinance also stems from a lack of grassroots participation in drafting the ordinance, among other factors. This study identifies a need for both perspectives to be critically examined and considered, especially considering that concerns raised from local narratives on the meaning-making of cultural practices may not be fully addressed due to limited grassroots participation in environmental governance.

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